Graphical Summary

1 · GWI model & datasets

BXD mice

CORT + DFP
(GWI-emulants)

RNA-seq · 8 days

dataset 1

320k methylation array · new

dataset 2

MBD-seq · 12 weeks

dataset 3

MBD-seq · 43 weeks

dataset 4

114 high-effect DMPs · 2,408 DE genes

≈ 3,000 methylation-QTL markers (LOD ≥ 4)

exposure time-course

gene lists · QTL markers

2 · Pinpoint — agentic LLM

Pinpoint · Claude Opus 4.8

Planner
query → ordered steps

Researcher
GeneNetwork
SPARQL

Expert
NCBI
Entrez

Reflector
critiques · redirects

Manager
routes · terminates

3 iterations → consensus answer
repeated sampling ×10 → robust signals

3 · Genomic tasks

QTL fine-mapping

GEMMA mQTL (LOD ≥ 4)
→ candidate genes in QTL regions

×10 repeats → 78 prioritized genes

Network analysis

genes across the 4 studies
→ weighted gene–gene links
(# shared annotations)

×10 repeats → 53 genes · 47 edges · 5 modules

network genes

PheWAS · BXD phenome

GeneNetwork · 13,700 traits · precomputed GEMMA

traits with a QTL signal at
network-gene loci (± 1 Mbp)

genes ↔ neurological & endocrine GWI-relevant traits

4 · Disease insights

5 gene modules converge on IL6–TNF signalling

Exposure
CORT + DFP (GWI-emulants)

Stress & cholinergic response
Fkbp5 · Hspa1a/b · Bche · Chrm3

Neuro-inflammation — IL6/TNF hub
Tlr2 · Cd14 · Tnf · Il6 · Il1b · Nlrp3 · Casp4

Oxidative stress & detoxification
Nqo1 · Hmox1 · Gsr · Txnrd1 · Pon1–3

Dysmyelination
Mbp · Mog · Plp1 ↓ → cognitive decline